MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 6754

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Rectifier, miniature, full wave

Outline --- See figure 1

Base --- E9-1

Envelope --- T6-1/2

Cathode --- Coated unipotential

Base connections:

Pin No.	---	1	2	3	4	5	6	7	8	9
Element	---	2a	nc	2k	h	h	nc	1k	nc	1a

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	epx	Epp/a	ib	Io	i(surge)	Rl	Cl	tk	Zp/a	(A) Ehk	TE	Alt
Unit:	V	v	Vac	ma	mAdc	a	Ohms	µF	sec	Ohms	v	°C	ft
Maximum:	6.6	1,450	Note 1	330	Note 1	1.10	---	---	45	Note 1	500	300	Note 1
Minimum:	6.0	---	---	---	---	---	---	---	---	---	---	---	---
TEST CONDITIONS:	6.3	---	350 (Note 2)	---	---	---	3,890	4	---	---	---	---	---

GENERAL:

Qualification - Required

Reliable tube

(A) denotes changes

Source: https://assist.dla.mil -- Downloaded: 202610-15T14:52Z
Check the source to verify that this is the current version before use.

METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1 (See note 3)							
1301	Heater current		0.65	II	If	0.9	1.1	A
1336	Heater-cathode leakage	Ehk = 500 Vdc (see note 5)	0.65	II	Ihk	---	50	µAdc
1353	Operation of rectifiers (1)	See note 4	0.65	II	Io	95	105	mAdc
1353	Operation of rectifiers (2)	Ef = 5.5 Vac	0.65	II	ΔIo_{E_f}	---	7	mAdc
1231	Emission	Eb = 15 Vdc (see note 5)	0.65	II	Is	65	---	mAdc
1201	Short and discontinuity detection		0.4	II	---	---	---	---
	Quality conformance inspection, part 2 (See note 3)							
1211	Insulation of electrodes	Ep-all = -1,500 Vdc (see notes 5 and 6)	2.5	S3	---	---	---	---
1031	Low-frequency vibration	10 G; F = 40 Hz; no voltages	6.5	Code H	---	---	---	---
1041	Shock	500 G; Epp/a = 0	---	---	---	---	---	---
1031	Vibration fatigue	No voltages	6.5	Note 7	---	---	---	---
---	Post-shock and vibration-fatigue test end points	Heater-cathode leakage Operation of rectifiers (1)	--- ---	--- ---	Ihk Io	--- 85	100 ---	µAdc mAdc
1121	Base strain		---	---	---	---	---	---
2126	Glass strain		2.5	I	---	---	---	---
(A) 1105	Permanence of marking		---	---	---	---	---	---

METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSP LEVEL OR CODE	ALLOWABLE DEFECTIVES PER CHARACTERISTIC 1ST SAMPLE	ALLOWABLE DEFECTIVES PER CHARACTERISTIC COMBINED SAMPLES	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 3									
1506	Heater-cycling life	Ef = 7.0 V; Ehk = 350 Vac; Eb = 0; 2-1/2 minutes on, 2-1/2 minutes off	---	---	---	---	---	3,000	---	Cycles
---	Heater-cycling life test end point	Heater-cathode leakage	---	---	---	---	Ihk	---	100	µAdc
1521	Survival-rate life	TA = room (see note 8)	---	---	---	---	---	---	---	---

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T14:52Z
Check the source to verify that this is the current version before use.

METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL PERCENT DEFECTIVE	INSP LEVEL OR CODE	ALLOWABLE DEFECTIVES PER CHARACTERISTIC		SYMBOL	LIMITS		UNIT
					SAMPLE	COMBINED SAMPLES		MIN	MAX	
	Quality conformance inspection, part 3 -Continued									
---	Survival-rate life-test end points	Short and discontinuity detection	---	---	---	---	---	---	---	---
		Operation of rectifiers (1)	---	---	---	---	Io	95	---	mAdc
---	Life-test (1) provisions	Survival-rate life-test conditions; TE = 300°C (see note 9)	---	---	---	---	---	---	---	---
---	Life-test (1) end points (500 hours)	Inoperatives	---	---	1	3	---	---	---	---
		Heater current	---	---	1	3	If	0.85	1.15	A
		Operation of rectifiers (1)	---	---	1	3	Io	90	---	mAdc
		Change in operation of rectifiers (1) of individual tubes	---	---	1	3	ΔIo t	---	8.0	mAdc
		Heater-cathode leakage	---	---	1	3	Ihk	---	100	μAdc
		Total defectives	---	---	2	4	---	---	---	---
---	Life-test (2) provisions	Survival-rate life-test conditions	---	Notes 10 & 11	---	---	---	---	---	---
---	Life-test (2) end points (500 hours)	Inoperatives	---	---	1	3	---	---	---	---
		Heater current	---	---	1	3	If	0.85	1.15	A
		Operation of rectifiers (1)	---	---	1	3	Io	90	---	mAdc
		Change in operation of rectifiers (1) of individual tubes	---	---	1	3	ΔIo t	---	8.0	mAdc
		Heater-cathode leakage	---	---	1	3	Ihk	---	100	μAdc
		Total defectives	---	---	2	4	---	---	---	---
---	Life-test (2) end points (1,000 hours)	Inoperatives	---	---	1	3	---	---	---	---
		Heater current	---	---	1	3	If	0.85	1.15	A
		Change in operation of rectifiers (1) of individual tubes	---	---	1	3	ΔIo t	---	10.0	mAdc
		Heater-cathode leakage	---	---	1	3	Ihk	---	100	μAdc
		Total defectives	---	---	3	6	---	---	---	---

NOTES:

1. To simplify the application of the maximum ratings to circuit design, absolute maximum ratings are presented in chart form as rating charts I, II, III, and IV. Operating points should be so selected that the boundary limits on rating charts I, II, III, and IV are not exceeded with a bogey tube under the worst probable operating conditions with respect to supply-voltage variation, equipment component variation, equipment control adjustment, and environmental conditions. A brief description of each of the rating charts is given below. The values of ac supply voltage as presented refer to the unloaded supply voltages per anode.

   (a) Rating chart I (see figure 2). This chart presents the maximum ratings for ac anode supply voltage and dc output current. The boundary FAEDG defines the limits for capacitor-input filter operation and the boundary FABCDG defines the limits for choke-input filter operation.

   (b) Rating chart II (see figure 3). This chart provides a convenient method for checking conformance with the maximum steady-state peak-anode-current rating. Rating chart II applies to capacitor-input filter operation only.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T14:52Z
Check the source to verify that this is the current version before use.

NOTES: - Continued

1. -Continued

   (c) Rating chart III (see figure 4). This chart shows the minimum value of anode-supply resistance (Rs) required to remain within the transient peak-anode-current rating. The value of Rs should be such that it lies to the left of the line on the rating chart at the highest probable value of line voltage. Rating chart III applies to capacitor-input filter operation only.

   (d) Rating chart IV (see figure 5). This chart presents the maximum ratings for ac anode supply voltage and altitude. Rating chart IV refers to both capacitor-input filter and choke-input filter operation.

2. All values of Epp/a refer to the unloaded supply voltage. The ratings refer to rectifier operation with sinusoidal supply voltages within the frequency range of 25 to 1,000 Hz.

(A) 3. Sampling procedure shall be in accordance with MIL-STD-105 with sample size determined by lot size, except the minimum sample size shall be as specified below. Use the AQL and inspection level specified for each individual test item to determine the minimum sample size code letter. The maximum lot tolerance percent defectives (LTPD), specified below, are for information purposes and are defined as the percent defective in the lot for which the probability of acceptance is 0.10 as obtained from the operating characteristic curves in MIL-STD-105.

AQL (percent defective)	Inspection level	Maximum LTPD	Minimum sample size code
0.4	II	2.7	L
0.65	II	3.3	L
2.5	I	12.9	H
2.5	S3	13.0	H

4. In a full-wave circuit, adjust Zp/a such that a tube having Etd = 15 Vdc at 85 mAdc per anode gives Io = 100 mAdc.

5. Test each section separately.

6. Heater and cathode tied together and positive with respect to anode.

(A) 7. This test shall be conducted on the initial lot and thereafter on a lot approximately every 6 months. When one lot has passed, the 6-month rule shall apply. In the event of lot failure, the lot shall be rejected and the succeeding lots shall be subjected to this test until a lot has passed. MIL-STD-105, sample size code letter E, shall apply.

(A) 8. In a full-wave life-test circuit, the values specified for Rl and C1 may be considered as approximate and shall be adjusted initially to give not less than Io = 85 mAdc with ib equal to or greater than 330 ma. Ehk = Eo.

(A) 9. Envelope temperature (TE) requirements, when measured in accordance with the temperature by conduction-band measurement (method 1226), will be satisfied if a tube having bogey Etd (+10 percent at Is = 46 mAdc per anode)under normal test conditions, is determined to operate at or above minimum specified temperature at any position in the life-test rack.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T14:52Z
Check the source to verify that this is the current version before use.

NOTES: - Continued

10. A 20-tube sample shall be selected from a lot at random in such a manner as to be representative of the lot. If such selection results in a sample containing tubes which are outside the initial specification sheet limits for the relevant life-test end-point characteristics, such tubes shall be replaced by randomly selected acceptable tubes.

11. In the event of failure of the first sample on life-test (2), select a completely fresh sample of 40 tubes which meet the acceptance inspection limits for those characteristics, specified as life-test (2) end points.

Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Army - EL
Navy -
Air Force - 11, 80
DSA - ES

User activities:
Army - WC
Navy - AS, OS, CG, SH
Air Force - 19

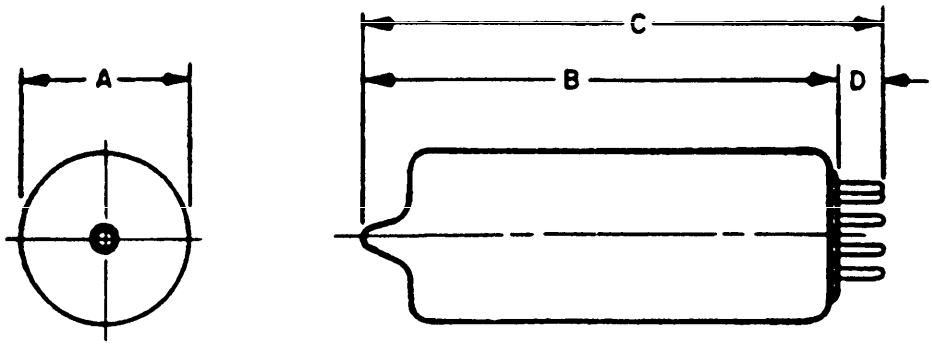
Preparing activity: Navy - EC

Agent: DSA - ES

(Project 5960-2425-83)

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T14:52Z
Check the source to verify that this is the current version before use.

truncated ruffle cone. The cone shall be 1.562 inches (39.67



Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 1		
A		.875 (22.23)
B		2.500 (63.50)
C		2.750 (69.85)
D	.250 (6.35) nom	

FIGURE 1. Outline drawing of electron tube type 6754.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T14:52Z
Check the source to verify that this is the current version before use.

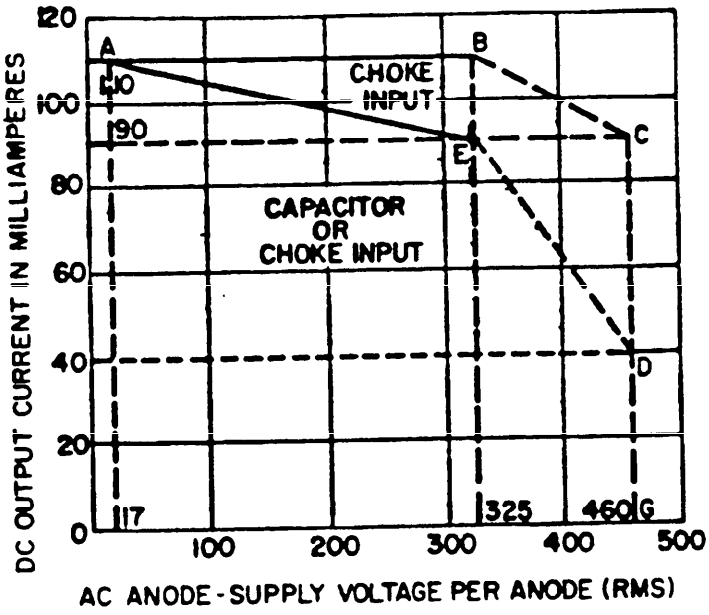


FIGURE 2. Rating chart I absolute maximum ratings.

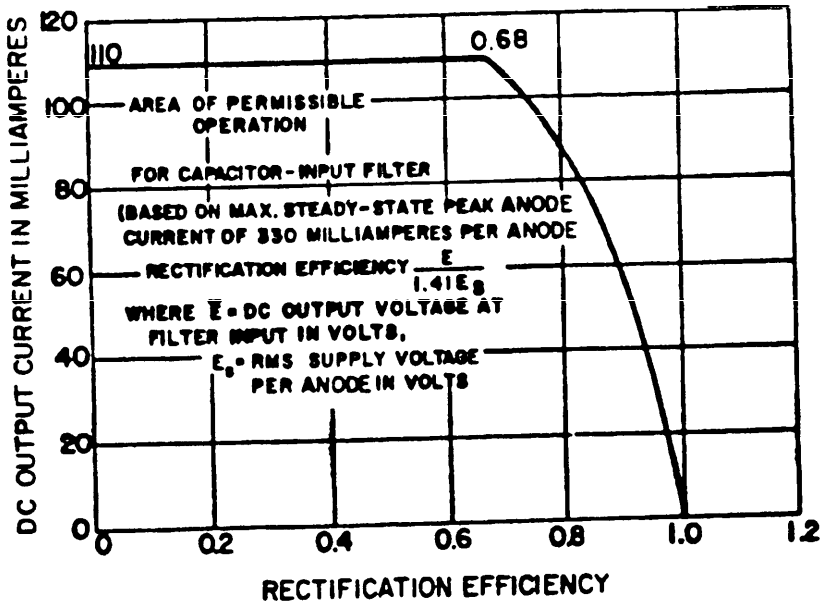


FIGURE 3. Rating chart II absolute maximum ratings.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T14:52Z
Check the source to verify that this is the current version before use.

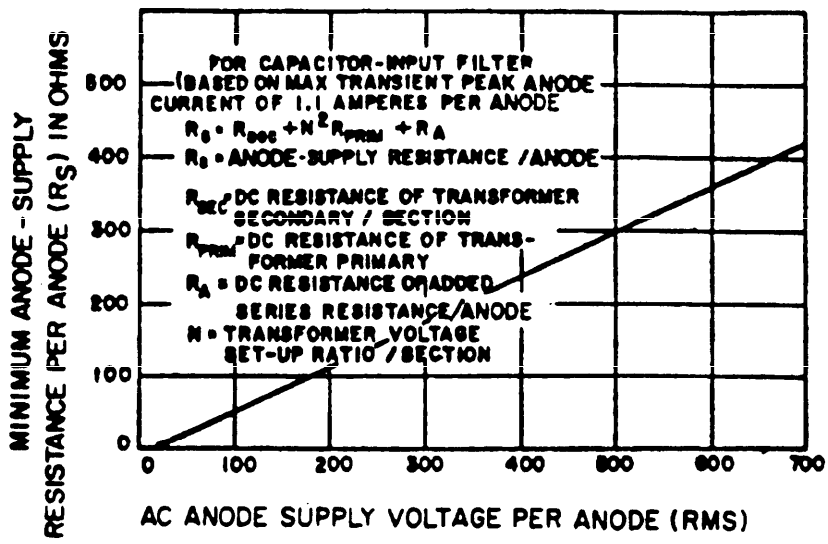


FIGURE 4. Rating chart III absolute maximum ratings.

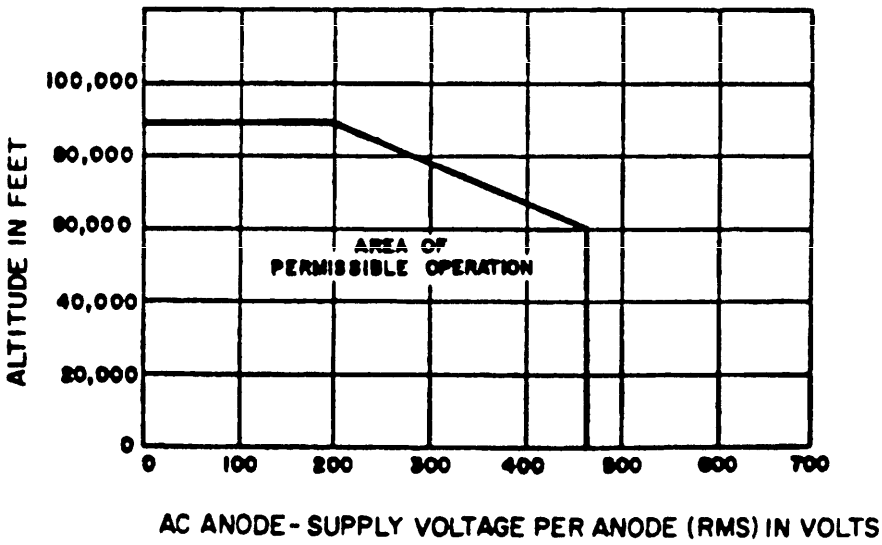


FIGURE 5. Rating chart IV absolute maximum ratings.

Source: https://assist.dla.mil -- Downloaded: 202610-15T14:52Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE NAVY
NAVAL ELECTRONIC SYSTEMS COMMAND
WASHINGTON, D. C. 20360

POSTAGE AND FEES PAID
NAVY DEPARTMENT

OFFICIAL BUSINESS

COMMANDER
NAVAL ELECTRONIC SYSTEMS COMMAND
DEFENSE STANDARDIZATION PROGRAM BRANCH
DEPARTMENT OF THE NAVY
WASHINGTON, D. C. 20360

FOLD

Source: https://assist.dla.mil -- Downloaded: 202610-15T14:52Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

S/N-0102-014-1801 C-25254

Source: https://assist.dla.mil -- Downloaded: 202610-15T14:52Z
Check the source to verify that this is the current version before use.